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Research Statement

My research is in econometrics and focuses on the credibility of empirical evidence. I study how empirical conclusions are affected by specification choices, researcher discretion, and limits to external validity, and how these degrees of freedom and sources of uncertainty can be formally measured or incorporated into decision-making. The motivation of my work is to start from problems that arise in empirical practice and connect them to tools from econometric theory, partial identification, and statistical decision theory. I start from judgments that researchers already make intuitively, and then build a formal measure from them: the formality makes such judgments more transparent and clarifies when intuition might be misguided.

My job market paper, *How Robust Are Robustness Checks?*, starts from a routine practice of empirical research and gives it a formal statistical treatment. Researchers commonly report alternative specifications and assess robustness by comparing their estimates with a preferred specification. In a survey of *AEJ: Applied* articles with complete replication packages, 177 of 179 report at least one robustness check, and the median paper reports about five per headline specification. Yet there is no standard statistical procedure for quantifying what one can learn from robustness checks, and the conclusions are often qualitative: estimates are "similar," a choice has "little impact." Simply comparing point estimates ignores both sampling uncertainty and the correlation between specifications, which can be substantial for checks estimated using the same data (the most common case).

I propose the robustness tolerance, the smallest distance b at which the hypothesis that every robustness-check estimand lies within b of the main-specification estimand is not rejected. The framework takes a preferred specification as given and robustness as the null: the tolerance is a lower confidence bound for the maximal discrepancy between the main estimand and its checks, obtained by inverting a moment-inequality test (Cox and Shi, 2023), and it accounts for both sampling uncertainty and the correlation of the estimates. Taking robustness as the null is what protects the author. A check whose estimand lies within the discrepancies already established by the other checks cannot push the measure above the true maximal discrepancy, except with probability α, however imprecise the check is, while a check whose estimand lies outside moves the tolerance once its discrepancy is large relative to its own standard error. No measure that takes fragility as the null has the first property, and neither does the maximal estimated discrepancy. For linear regressions I provide a plug-in covariance estimator for the stacked coefficients under which the procedure is uniformly valid.

I also develop relative and standardized versions of the tolerance, which express discrepancies in units of the main effect and in units of the variation in the outcome and treatment. In an application to the wage-setting results of Massenkoff and Wilmers (2023), a check the authors single out as an exception moves the point estimate by 79% of the main effect; the relative tolerance shows the estimand moves by at least 9%, because the discrepancy is less than three standard errors from zero. For all of the authors' explicit robustness checks, the tolerance is smaller than the naive discrepancy between point estimates, and the difference depends on how precisely each discrepancy is estimated.

The paper also connects robustness checks to formal sensitivity analysis. The robustness tolerance is not a measure of omitted variable bias: it is instead a measure of the bias from observed specification choices. When those choices can be related to a sensitivity model, however, the tolerance gives a benchmark for calibrating the sensitivity parameter. I work this out for the parameter of Diegert, Masten, and Poirier (2026). If a researcher requires the sensitivity model to allow at least as much unobserved bias as the bias observed in robustness checks, the tolerance becomes a lower bound on the sensitivity parameter. It also says how much a set of robustness checks has actually explored, in the same units the sensitivity literature uses for unobservables.

My current project, *Detecting Robustness Hacking*, turns from the interpretation of robustness checks to evaluating the process that generates them. The robustness tolerance takes the reported checks as

given and measures how much they agree with the main specification, but it cannot stop a researcher from running many checks and reporting only those that agree. I ask whether this happens and whether it is detectable. The problem is different from p-hacking: there is no clear threshold, since a check is made to look "close" to the main estimate, and "close" is not judged formally. And the goal is usually agreement over significance, so there would still be an incentive to hack a check that is significant but far from the main estimate. The existing tests for p-hacking assume a cutoff and a one-directional incentive, so they do not immediately translate to this context.

The empirical part of the project is a replication census of the *American Economic Journal: Applied Economics* from 2011 to 2026, covering the papers with full data availability in Paul Goldsmith-Pinkham's replication-package database. Large language model agents, following a protocol I designed, re-estimate every main specification and robustness check in each paper from the deposited data and code, check each estimate against the published table, and compute the joint covariance of all the estimates with the plug-in estimator from my job market paper. I expect to have around 140 fully replicated papers. I look at the p-value of the Wald test that all the reported checks agree with the main specification, pooled across papers. If nothing was hacked, this p-curve has to be non-increasing, and for Wald tests I show that it also satisfies additional shape restrictions; both hold no matter how the true discrepancies are distributed. I then consider a few explicit rules a researcher might follow when choosing what to report and ask which of them the p-curve restrictions could detect. The answer, even in theory, depends on the true distribution of the checks and on the hacking rule employed. Dropped checks also leave a trace in the distribution of the reported discrepancies, which has a gap right above whatever cutoff the researcher used, and I test for that gap with a density discontinuity test. All of these tests need the covariance between each check and the main estimate, which no published table reports. This is why the census re-estimates every paper from its deposited data instead of reading the published tables, and why it needs language-model agents to do so at scale. This calls for making the reporting of covariance matrices routine. A draft is available upon request.

A third project turns to a condition the researcher does not control, the external validity of experimental evidence. In *Externally Valid Selection of Experimental Sites via the k-Median Problem*, joint with José Luis Montiel Olea, Chen Qiu, Jörg Stoye, and Yiwei Sun, we study how to choose where to conduct experiments when the objective is to inform treatment decisions in other populations. We formulate the choice of experimental sites as a statistical decision problem in which treatment effects may differ between experimental and policy-relevant sites. When treatment-effect heterogeneity can be substantial, we show conditions under which the minimax-regret optimal selection of experimental sites is equivalent to solving a k-median problem, a classical problem in computer science and operations research. The equivalence allows a difficult experimental-design problem to be solved with existing algorithms for k-median. The paper is under revision at the *Journal of Political Economy: Microeconomics*, and an extended abstract appeared in the Proceedings of the 26th ACM Conference on Economics and Computation.

These projects lead to several directions for my future research, most of which expand the robustness agenda. First, I want to move from evaluating a given set of robustness checks to studying how they should be chosen in the first place. One possibility is to formulate their design as a statistical decision problem: given the empirical choices about which a researcher is uncertain, which checks are most informative about the credibility of the conclusion? Second, the replication census provides what simulations cannot: the configurations and correlations of real robustness checks. I plan to compare the power of the moment-inequality tests that could be inverted to compute the tolerance, including the test of Cox and Shi (2023), on the replicated papers rather than on simulated designs, as an appendix to the job market paper or as a short companion paper. Third, the census also yields the tolerance for every replicated paper, which will make it possible to describe how robust published results in a leading applied journal actually are, in comparable units, and whether verbal robustness claims track measured ones; an R package implementing the tolerance will be released in the near future. Finally, I plan to build on the use of AI in my robustness-hacking project to study its role in the production and verification of empirical research. Across these projects, I aim to develop econometric tools that make the sources of uncertainty in empirical research explicit and help researchers draw conclusions that are commensurate with the evidence available to them.